

INTERVIEW PREP

SQL QUESTIONS FOR DATA ANALYST ROLES

VARUN SAGAR THEEGALA

SQL IS EVERYWHERE

If you're about to appear for a Data Analyst interview, you're certainly going to be grilled in SQL.

SQL is at the core of any data analyst's job.

So **definitely** interviewers will **test your foundations** in the database language.

Hence, it's **crucial to know** what are the **concepts** you'll more **certainly encounter** in these situations.

**SO HERE ARE 25
QUESTIONS YOU
SHOULD BE
PREPARED
FOR →**

(With The Answers)

1. What is DDL and DML? What is the difference?

ANSWER ->

- DDL (Data Definition Language) is used to define the structure of the database (e.g., CREATE TABLE, ALTER TABLE, DROP TABLE).
- DML (Data Manipulation Language) is used to manipulate the data within the database (e.g., INSERT, UPDATE, DELETE, SELECT).

2. What is the difference between Truncate vs Delete vs Drop?

ANSWER ->

- Truncate removes all rows from a table quickly. It cannot be rolled back.
- Delete removes specific rows from a table based on a condition. It can be rolled back.
- Drop removes the entire table and its structure from the database. It cannot be rolled back.

3. When is Group By used?

ANSWER ->

The GROUP BY clause is used with aggregate functions (like SUM, AVG, COUNT) to group rows based on one or more columns and perform calculations within each group.

4. What is the difference between Where clause vs. Having clause?

ANSWER ->

- Where clause filters rows before grouping and aggregation.
- Having clause filters groups after grouping and aggregation.

5. What are Scalar and Aggregate functions in SQL?

ANSWER ->

- Scalar functions return a single value for each row (e.g., UPPER, LOWER, SUBSTRING).
- Aggregate functions return a single value for a group of rows (e.g., SUM, AVG, COUNT, MAX, MIN).

6. What does coalesce function do in SQL?

ANSWER ->

The COALESCE function returns the first non-null value in a list of arguments.

7. What are Constraints? How many Constraints are there?

ANSWER ->

- Constraints are rules that enforce data integrity in a database.
- Some common constraints include PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK.

8. How many primary key and unique key can be in a table?

ANSWER ->

- A table can have only one PRIMARY KEY constraint.
- A table can have multiple UNIQUE constraints.

9. What are Joins in SQL. Types of Joins in SQL?

ANSWER ->

- Joins combine data from two or more tables based on a related column.
- Common types include INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN, and CROSS JOIN.

10. Difference between UNION and UNION ALL

ANSWER ->

- UNION combines the results of two or more SELECT statements, removing duplicate rows.
- UNION ALL combines the results of two or more SELECT statements, including duplicate rows.

11. What is the order of execution in SQL Code?

ANSWER ->

The general order of execution is: FROM, WHERE, GROUP BY, HAVING, SELECT, ORDER BY.

12. How many Joins is required to join 8 tables?

ANSWER ->

To join 8 tables, you would typically require 7 joins.

13. What are Window Functions or Analytical functions?

ANSWER ->

Window functions perform calculations across a set of rows related to the current row, such as ranking, partitioning, and calculating running totals.

14. What is the diff between Rank & Dense Rank Function?

ANSWER ->

- Rank assigns a rank to each row within a partition, with gaps for ties.
- Dense Rank assigns a rank to each row within a partition, without gaps for ties.

15. What is View? How it is different from Table?

ANSWER ->

- A view is a virtual table that is created from the result of an SQL query.
- It doesn't physically store data but provides a dynamic view of the underlying data.

16. What is Index?

ANSWER ->

- An index is a data structure that improves the speed of data retrieval by creating a sorted order for specific columns in a table.

17. What is CTE? How it is useful?

ANSWER ->

- A Common Table Expression (CTE) is a temporary named result set that you can define within a SELECT, INSERT, UPDATE, or DELETE statement.
- CTEs can make complex queries more readable and maintainable.

18. What is difference between SQL vs. PLSQL?

ANSWER ->

- **SQL (Structured Query Language)** is used to interact with databases, primarily for data retrieval and manipulation.
- **PL/SQL (Procedural Language/SQL)** is an extension of SQL that adds procedural programming capabilities to SQL, such as loops, conditional statements, and variables.

19. What is the use case of Cross Join?

ANSWER ->

- CROSS JOIN is used to create a Cartesian product (all possible combinations) of rows from two tables.
- It is often used in conjunction with other clauses (like WHERE) to filter the results.

20. Explain the ACID properties of transactions and their importance.

ANSWER ->

ACID stands for Atomicity, Consistency, Isolation, and Durability. These properties ensure the reliability and integrity of database transactions.

- **Atomicity:** All-or-nothing execution of a transaction.
- **Consistency:** Maintains database integrity after a transaction.
- **Isolation:** Transactions are independent and do not interfere with each other.
- **Durability:** Committed changes are permanent and survive system failures.

21. Describe the difference between a Primary & Foreign Key.

ANSWER ->

- **Primary Key:** A unique identifier for each row in a table. It cannot contain null values and must be unique for every record.
- **Foreign Key:** A column in one table that references the Primary Key of another table. It establishes a relationship between the two tables.

22. How do Indexes work in a database?

ANSWER ->

Indexing stores a sorted list of values for one or more columns, along with pointers to the actual data rows.

Benefits:

- Significantly improves query performance, especially for searches, joins, and sorting.
- Enables faster data retrieval and updates.

Drawbacks:

- Increased storage space required for the index.
- Can slow down data insertion and updates as the index needs to be maintained.

23. Explain the concept of normalization in database design.

ANSWER ->

Normalization is the process of organizing data in a database to minimize redundancy and improve data integrity. It involves breaking down large tables into smaller, more manageable tables with well-defined relationships.

Why is it important:

- Improves data consistency and reduces the risk of data anomalies.
- Makes data easier to maintain and update.
- Improves query performance.

24. Explain Subqueries & its different types with examples.

ANSWER ->

Subqueries are nested SELECT statements within another SQL statement.

- **Scalar Subquery:** Returns a single value (e.g., used in WHERE clause to compare a column value with the result of the subquery).
- **Correlated Subquery:** References a column from the outer query within the subquery.
- **Multiple-Row Subquery:** Returns multiple rows (e.g., used with IN or EXISTS operators).

25. Explain left join, right join, and full outer join with examples.

ANSWER ->

- **LEFT JOIN:** Returns all rows from the left table and matching rows from the right table. If there is no match in the right table, NULL values are returned for the right table's columns.
- **RIGHT JOIN:** Returns all rows from the right table and matching rows from the left table. If there is no match in the left table, NULL values are returned for the left table's columns.
- **FULL OUTER JOIN:** Returns all rows from both tables. If there is no match in either table, NULL values are returned for the missing columns.

CONCLUSION

A strong foundation in SQL theory is **not** just about **answering interview questions**.

It's about demonstrating your **understanding of core database concepts** and your **ability to think critically** and solve complex problems.

By mastering these fundamentals, you'll be well-equipped to **tackle real-world data challenges** and excel as a data analyst.